Online Auction System

Problem Statement

In general, the traditional auction system demands the physical presence of the user. With the proposed **Online Auction System**, the facility to sell items or take part in auctions **through the internet will be provided to the user in an efficient manner**.
In the proposed system, the facility to register the users, list the items for auction, conduct the auctions with the facility to track the bids, conclude the auctions with the identification of the winning user, etc., will be provided.
This project will be an example for the implementation of object-oriented programming techniques, modular programming, etc., along with the usage of various features of the Java programming language.

Target User

- Individuals who want to **sell items through auctions**
- Buyers interested in **bidding on products online**
- System administrators who manage auction listings and users

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Core Features

- User registration and login system
- Item listing for auction with starting price and time limit
- Real-time bidding on available items
- Automatic identification of the **highest bidder**
- Auction timer and automatic auction closing
- Bid history tracking for each item
- Winner notification after auction completion
- Data storage for users, items, and bids

OOP Concepts to be Used

Abstraction: It will define general behaviours like auction operations and user operations.
Inheritance: Various types of users like Buyer, Seller, Admin, etc., can be inherited from a parent class named User.
Polymorphism: Operations like bidding and viewing item information can be performed.
Exception Handling: It will handle exceptions like invalid bids, time expiration of the auction, and incorrect user information.

Proposed Architecture Description

It will be a system with a **modular layered architecture** with the application split into modules like **User Management, Auction Management**, etc. Each module will be having classes to take care of the operations. It will also be having the information about the users, the items for auctioning, the bids stored through **Java Collections and file handling operations**. It will be a system with a focus on **encapsulation, class hierarchy, and component reuse** to ensure the application logic, the data handling, the user interactions are well organized within the system.